



Java Project Guidline

Package Fix

DTO



```
1 package dto;
```

DAO



```
1 package dao;
2 import dto.LibraryDTO;
```

Service



```
1 package service;
2 import dao.*;
3 import dto.LibraryDTO;
```

MainApp (optional)



```
1 package ui; // or remove later
```

First GUI Window

Create ui/LibraryUI.java:



```
1  package ui;
2
3  import javax.swing.*;
4  import service.LibraryService;
5
6  public class LibraryUI {
7
8      public static void main(String[] args) {
9
10         JFrame frame = new JFrame("Library Management System");
11         frame.setSize(500, 400);
12         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
13         frame.setLayout(null);
14
15         JLabel heading = new JLabel("Library Management System");
16         heading.setBounds(140, 20, 250, 30);
17
18         JButton addBtn = new JButton("Add Book");
19         addBtn.setBounds(150, 80, 200, 30);
20
21         JButton viewBtn = new JButton("View Books");
22         viewBtn.setBounds(150, 130, 200, 30);
23
24         JButton deleteBtn = new JButton("Delete Book");
25         deleteBtn.setBounds(150, 180, 200, 30);
26
27         frame.add(heading);
28         frame.add(addBtn);
29         frame.add(viewBtn);
30         frame.add(deleteBtn);
31
32         frame.setVisible(true);
33
34         LibraryService service = new LibraryService();
35
36         addBtn.addActionListener(e -> {
37
38             JFrame addFrame = new JFrame("Add Book");
39             addFrame.setSize(400, 300);
40             addFrame.setLayout(null);
41
42             JLabel t = new JLabel("Title:");
43             t.setBounds(50, 50, 100, 30);
44             JTextField titleField = new JTextField();
45             titleField.setBounds(150, 50, 150, 30);
46
47             JLabel a = new JLabel("Author:");
48             a.setBounds(50, 100, 100, 30);
49             JTextField authorField = new JTextField();
50             authorField.setBounds(150, 100, 150, 30);
51
52             JLabel q = new JLabel("Quantity:");
53             q.setBounds(50, 150, 100, 30);
54             JTextField qtyField = new JTextField();
55             qtyField.setBounds(150, 150, 150, 30);
56
57             JButton submit = new JButton("Submit");
58             submit.setBounds(150, 200, 100, 30);
59
60             addFrame.add(t);
61             addFrame.add(titleField);
62             addFrame.add(a);
63             addFrame.add(authorField);
```

```

64         addFrame.add(q);
65         addFrame.add(qtyField);
66         addFrame.add(submit);
67
68         addFrame.setVisible(true);
69
70         submit.addActionListener(ev -> {
71             try {
72                 String title = titleField.getText();
73                 String author = authorField.getText();
74                 int qty = Integer.parseInt(qtyField.getText());
75
76                 service.addBook(new dto.LibraryDTO(0, title, author, qty));
77
78                 JOptionPane.showMessageDialog(addFrame, "Book Added!");
79                 addFrame.dispose();
80
81             } catch (Exception ex) {
82                 JOptionPane.showMessageDialog(addFrame, "Invalid Input!");
83             }
84         });
85     });
86
87     viewBtn.addActionListener(e -> {
88
89         JFrame viewFrame = new JFrame("All Books");
90         viewFrame.setSize(600, 400);
91
92         String[] columns = {"ID", "Title", "Author", "Quantity"};
93
94         java.util.List<dto.LibraryDTO> books = service.getAllBooks();
95
96         String[][] data = new String[books.size()][4];
97
98         for (int i = 0; i < books.size(); i++) {
99             data[i][0] = String.valueOf(books.get(i).getId());
100             data[i][1] = books.get(i).getTitle();
101             data[i][2] = books.get(i).getAuthor();
102             data[i][3] = String.valueOf(books.get(i).getQuantity());
103         }
104
105         JTable table = new JTable(data, columns);
106         JScrollPane sp = new JScrollPane(table);
107
108         viewFrame.add(sp);
109         viewFrame.setVisible(true);
110     });
111
112     deleteBtn.addActionListener(e -> {
113
114         String input = JOptionPane.showInputDialog("Enter Book ID:");
115
116         try {
117             int id = Integer.parseInt(input);
118             service.deleteBook(id);
119
120             JOptionPane.showMessageDialog(null, "Book Deleted!");
121
122         } catch (Exception ex) {
123             JOptionPane.showMessageDialog(null, "Invalid ID!");
124         }
125     });
126 }
127 }

```

Commands to Compile and Run Java Programs

Run these all from the root dir of the project, not src

```
>_
```

```
1 $ find . -name "*.class" -delete
```

```
>_
```

```
1 $ tree.com //f //a
2 Folder PATH listing for volume Windows-SSD
3 Volume serial number is 3036-E42B
4 C:.\
5 |   .classpath
6 |   .gitignore
7 |   .project
8 |   README.md
9 |
10 \---src
11 |   MainApp.java
12 |   mysql-connector-j-9.7.0.jar
13 |
14 +---dao
15 |   LibraryDAO.java
16 |   LibraryDAOImpl.java
17 |
18 +---dto
19 |   LibraryDTO.java
20 |
21 +---service
22 |   LibraryService.java
23 |
24 \---ui
25     LibraryUI.java
```

```
>_
```

```
1 $ javac -d . -cp src/mysql-connector-j-9.7.0.jar src/**/*.java
```



```
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10 +---dao
11 |       LibraryDAO.class
12 |       LibraryDAOImpl.class
13 |
14 +---dto
15 |       LibraryDTO.class
16 |
17 +---service
18 |       LibraryService.class
19 |
20 +---src
21 |   |   MainApp.java
22 |   |   mysql-connector-j-9.7.0.jar
23 |   |
24 |   +---dao
25 |   |       LibraryDAO.java
26 |   |       LibraryDAOImpl.java
27 |   |
28 |   +---dto
29 |   |       LibraryDTO.java
30 |   |
31 |   +---service
32 |   |       LibraryService.java
33 |   |
34 |   \---ui
35 |           LibraryUI.java
36 |
37 \---ui
38         LibraryUI.class
```



```
1 $ java -cp ".;src/mysql-connector-j-9.7.0.jar" ui.LibraryUI
```

MySQL Connector

- Download it from:  <https://dev.mysql.com/downloads/connector/j/>
(<https://dev.mysql.com/downloads/connector/j/>).
- Choose:
 - Platform Independent
 - Download ZIP
- Extract: mysql-connector-j-9.7.0.jar
- Move it here:

```
1 Library-Management-System/
2 |─ src/
3 |─ mysql-connector-j-9.7.0.jar  📁 HERE
```

MySQL Installer

- Download it from:  <https://dev.mysql.com/downloads/installer/>
(<https://dev.mysql.com/downloads/installer/>).
- Create Database



```
1 CREATE DATABASE rnsitdb;  
2 USE rnsitdb;
```

- Create Table



```
1 CREATE TABLE books (  
2     id INT PRIMARY KEY AUTO_INCREMENT,  
3     title VARCHAR(100),  
4     author VARCHAR(100),  
5     quantity INT  
6 );
```

- Update Your Java Code



```
1 private static final String URL = "jdbc:mysql://localhost:3306/rnsitdb";  
2 private static final String USER = "root";  
3 private static final String PASS = "your_password";
```